
Statistics 212: Lecture 1 (January 27, 2025)

Preview of Topics and Radon-Nikodym Theorem

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1 Lecture 1

Today we're focusing on a preview of future topics and proof of Radon-Nikodym Theorem. Main topics for today:

- Advanced martingales
- Brownian motion
- Ito (Stochastic) Calculus

See [Instructor Website](#) for more info. Also, sign ups for 5 minute meetings with Mark Sellke 1-2:30 Wed, Jan 29 or Mon Feb 3. Form to be sent out!

1.1 Preview of Brownian Motion and Ito Calculus

Definition 1.1 (Brownian Motion).

- (a) Einstein's definition. Scaling limit of a simple random walk. An example of a simple random walk is

$$\begin{aligned}x_0 &= 0 \\x_1 &= \pm 1 \\x_2 &= x_1 \pm 1 \\&\vdots\end{aligned}$$

where all the $\pm$ are iid uniform. If we “scale out” the graph of the simple random walk, by the central limit theorem, we have $x_t \approx \mathcal{N}(0, t)$ where t is a very large number (say a googol). We have $B_s = \frac{x_t}{\sqrt{10^{100}}} \sim \mathcal{N}(0, s)$. Graphing out these B's, we obtain a graph that is a random fractal.

- (b) Wiener's definition. Brownian motion on $t \in [0, 1]$ is a random Fourier series. We have

$$B_t = g_0 t + \sum_{k \geq 1} g_k \sqrt{2} \frac{\sin(\pi k t)}{\pi k},$$

where $g_0, g_1, \dots$ are IID standard Gaussian. (This is called a Karhunen–Loève decomposition.)

(c) Gaussian process. The value at every timestamp is a Gaussian. We have

$$E[B_s] = 0$$

$$\text{Cov}(B_{s_1}, B_{s_2}) = \min(s_1, s_2).$$

Explanation for covariance: If $s_1 < s_2$, then

$$E[B_{s_1}^2] = s_1$$

$$E[B_{s_1}(B_{s_2} - B_{s_1})] = 0,$$

so $E[B_{s_1}B_{s_2}] = s_1$.

Definition 1.2 (Ito Calculus). Can think about Ito Calculus as calculus for Brownian motion or processes that are similar to Brownian motion.

Examples:

- Stock prices. They are continuous, and we can think of it as a martingale. However, a stock price can have a time-changing volatility, which is not quite Brownian (Brownian motion has constant volatility over time, so it looks the same everywhere). Ito Calculus allows us to analyze these quasi-Brownian objects.
- $Z_t = B_t^2$. The process would clearly never go negative, and it oscillates more when at large values. Even though this isn't quite Brownian, we can still use Ito calculus on this process.
- Used for biology, optimal control, PDEs, complex analysis, diffusion sampling, quantum mechanics, etc.

1.2 Radon-Nikodym Theorem

Definition 1.3 (Absolute continuity of finite measures). $\nu \ll \mu$ indicates that the finite measure ν is **absolutely continuous** with respect to μ . This means that for every measurable set S such that $\mu(S) = 0$, we also have $\nu(S) = 0$.

Theorem 1.4 (Radon-Nikodym Theorem). *Start off with finite measure μ on $(\Omega, \mathcal{F})$. Essentially, RN tells us what kind of measures we can produce starting with μ .*

- (a) *If $\nu \ll \mu$ (i.e., ν is absolutely continuous with respect to μ), then there exists a non-negative integrable f such that*

$$\nu(S) = \int_S f(\omega) d\mu(\omega) = \int_S f d\mu$$

for any measurable set $S \in \mathcal{F}$. We define $f = \frac{d\nu}{d\mu}$ as the Radon-Nikodym derivative.

- (b) *(More general) Without assuming absolute continuity, there exists a non-negative integrable f and finite measure Θ such that we can decompose*

$$\nu(S) = \Theta(S) + \int_S f(\omega) d\mu(\omega).$$

Furthermore Θ, μ are disjointly supported, i.e. there exists an $S \in \mathcal{F}$ such that $\mu(S) = 0$ and $\Theta(\Omega \setminus S) = 0$. We call $\Theta(S)$ the “singular part” and the second term the “absolute continuous part” of ν .

- (c) *Assume $\nu \leq \mu$, i.e. $\nu(S) \leq \mu(S)$ for all $S \in \mathcal{F}$. Then there exists a measurable $f : \Omega \rightarrow [0, 1]$ with $f = d\nu/d\mu$.*

Remark. We can show $(b) \implies (a) \implies (c)$. Also note that if we're given (c) , then for general finite measures (ν, μ) , we have $\nu \leq \nu + \mu$ so we can simply apply (c) to the pair $(\nu, \nu + \mu)$. On the first homework, we will show this recovers (a) and (b) . Intuitively, all 3 statements have the same core difficulty, that one has to “conjure up” the function f out of thin air.

Proof. Simplest proof that Mark was able to find, by Anton Schep (2003). We prove the RN Theorem in the form (c) . The idea is to find the largest f such that $f \leq d\nu/d\mu$ and show equality holds. Define

$$H = \left\{ f : \Omega \rightarrow [0, 1]; \forall S \in \mathcal{F}, \int_S f d\mu \leq \nu(S) \right\}. \quad (1)$$

We want to find the maximum of H . For intuition, one can consider what happens **assuming** a Radon-Nikodym derivative $f_* = \frac{d\nu}{d\mu}$ exists. Then for arbitrary measurable f_1 , we have $f_1 \in H$ if and only if $f_1 \leq f_*$ holds almost everywhere.

Our first claim that H is closed under maximum, i.e. if $f_1, f_2 \in H$ then $\max(f_1, f_2) \in H$. Assuming a Radon-Nikodym derivative exists, this is just because if $f_1, f_2 \leq f_*$ almost everywhere, then $\max(f_1, f_2) \leq f_*$. However we can prove it just from the given condition. Let $A = \{\omega \in \Omega : f_1 \geq f_2\}$ and $B = \{\omega \in \Omega : f_1 < f_2\}$ (the complement of A). We have

$$\begin{aligned} \int_S \max(f_1, f_2) d\mu &= \int_{S \cap A} f_1 d\mu + \int_{S \cap B} f_2 d\mu \\ &\leq \nu(S \cap A) + \nu(S \cap B) \\ &= \nu(S). \end{aligned}$$

Thus, $\max(f_1, f_2) \in H$ as well.

Following this observation, we will aim to demonstrate $f_* \in H$ by taking repeated maximums. We attempt to define $f_*(\omega) = \max_{f \in H} f(\omega)$. But this is a faulty definition. Suppose that $\mu(\{\omega\}) = 0 \forall \omega \in \Omega$, i.e. μ has no atoms. Then $f_*(\omega) = 1$ for all ω because $f_\omega(x) = \mathbb{1}_{x=\omega} \in H$.

Instead, we have to take the max of finitely or countably many functions. For $k = 1, 2, \dots$, define $g_k : \Omega \rightarrow [0, 1]$. as follows. Let

$$M = \sup_{f \in H} \int_\Omega f d\mu.$$

We require $g_k \in H$ with $\int_\Omega g_k d\mu \geq M - \frac{1}{k}$. We can repeatedly take maximums as so:

$$\begin{aligned} f_1 &= g_1 \in H \\ f_2 &= \max(g_1, g_2) \in H \\ f_3 &= \max(g_1, g_2, g_3) = \max(f_2, g_3) \in H. \\ &\vdots \end{aligned}$$

Note that $0 \leq f_1 \leq f_2 \leq \dots \leq 1$. By the monotone convergence theorem, there exists an $f_* = \lim_{k \rightarrow \infty} f_k$. We want to show that f_* is the RN-derivative.

We can see $f_* : \Omega \rightarrow [0, 1]$. Less obvious but crucial is that

$$\int_\Omega f_* d\mu = M. \quad (2)$$

As

$$\int_\Omega f_* d\mu \geq \int_\Omega g_k d\mu \geq M - \frac{1}{k}$$

for all $k \in \mathbb{N}$, we see $\int_{\Omega} f_* d\mu \geq M$. By Fatou's Lemma, we see $\int_{\Omega} f_* d\mu \leq M$ as $f_k \in H$ for all k . We also see that $f_* \in H$ as by Fatou's Lemma, we have $\int_S f_* d\mu \leq \liminf_{k \rightarrow \infty} \int_S f_k d\mu \leq \nu(S)$.

To show that $\int_S f_* d\mu = \nu(S)$ for all S , we can proceed with proof by contradiction. Assume that there exists some S such that $\int_S f_* d\mu < \nu(S)$, so intuitively there is a “deficit” in S that we have not yet exhausted. We will try to increase f_* while remaining in H , which contradicts maximality of M (due to (2)).

Define $E_1 = \{\omega : f_*(\omega) = 1\}$. We first show that the “deficit” does not come from the part of S in E_1 . Recall the initial assumption that $\nu \geq \mu$. But we also have

$$\mu(S \cap E_1) = \int_{S \cap E_1} f_* d\mu \leq \nu(S \cap E_1)$$

as $f_* \in H$. Then

$$\mu(S \cap E_1) = \int_{S \cap E_1} f_* d\mu = \nu(S \cap E_1).$$

Therefore we can replace S by $S \setminus E_1 = S \cap E_0$, where $E_0 = \Omega \setminus E_1$ is the complement of E_1 .

Next we want a positive amount of space to increase f_* , so we “exhaust” E_0 . For each $n \geq 1$, define

$$F_n = \{\omega : f_*(\omega) \leq 1 - \frac{1}{n}\}.$$

These F_n exhaust E_0 in that $F_1 \subseteq F_2 \subseteq \dots$, and

$$\bigcup_{n=1}^{\infty} F_n = E_0.$$

Then $\int_{S \cap E_0} f_* d\mu < \nu(S \cap E_0)$, which implies $\int_{S \cap F_n} f_* d\mu < \nu(S \cap F_n)$ for large n . Define $\bar{S} = S \cap F_n$. For $\epsilon > 0$ sufficiently small, we have

$$\int_{\bar{S}} (f_* + \epsilon \chi_{\bar{S}}) d\mu < \nu(\bar{S}). \quad (3)$$

We want to show $f_* + \epsilon \chi_{\bar{S}} \in H$ to contradict the maximality of f_* . Although (3) appears like the condition to be within H , it is only for a specific set $\bar{S}$, while H is a condition on *all* measurable sets. So a bit more is still needed.

To finish the proof, we need one more exhaustion argument. First, the condition for $f_* + \epsilon \chi_{\bar{S}}$ to be in H holds on any set disjoint from $\bar{S}$, since the extra $\epsilon \chi_{\bar{S}}$ term doesn't matter. So it remains to handle subsets $\tilde{S} \subseteq \bar{S}$ (since in general we can decompose $\bar{S}$ into $\tilde{S} \cap \bar{S}$ and $\bar{S} \setminus \tilde{S}$). It will be convenient to define the “ ϵ -deficit”

$$\text{Def}_{\epsilon}(A) = \nu(A) - \int_A f_* d\mu - \epsilon \mu(A).$$

Note that this function is additive. Further, $\text{Def}_{\epsilon}(\bar{S}) > 0$ by (3). The idea is that if $S_1 \subseteq \bar{S}$ violates the H -condition, i.e.

$$\int_{S_1} (f_* + \epsilon \chi_{\bar{S}}) d\mu = \int_{S_1} (f_* + \epsilon \chi_{S_1}) d\mu > \nu(S_1),$$

this means $\text{Def}_{\epsilon}(S_1) < 0$ is negative. Then we can simply remove S_1 and use additivity of this functional to find that

$$\text{Def}_{\epsilon}(\bar{S} \setminus S_1) = \text{Def}_{\epsilon}(\bar{S}) - \text{Def}_{\epsilon}(S_1) \geq \text{Def}_{\epsilon}(\bar{S}) > 0. \quad (4)$$

Thus intuitively, removing a violating set S_1 (with negative deficit) only widens the deficit. So by removing “all possible violating sets”, there will be no more room for violations. To be precise, we construct a sequence of disjoint sets $S_1, S_2, \dots \subseteq \bar{S}$, each of which attains an “almost maximal violation” (subject to the restriction of disjointness with previous sets). Namely define

$$a_k = \inf_{S_k} \text{Def}_{\epsilon}(S_k)$$

with the infimum being over $S_k \subseteq \bar{S}$ disjoint with $S_1, S_2, \dots, S_{k-1}$. We have $a_k \leq 0$ since the empty set is always an option. We'll choose S_k to approximately optimize this infimum up to margin $1/k$, i.e.:

$$\text{Def}_\epsilon(S_k) \leq a_k + \frac{1}{k}. \quad (5)$$

Then we can define

$$\hat{S} = \bar{S} \setminus \left(\bigcup_{k \geq 1} S_k \right)$$

to be “ $\bar{S}$ with all the deficit removed”. Now let's verify that $\hat{f} = f + \epsilon \chi_{\hat{S}} \in H$, and that this contradicts maximality of M to finish the proof:

- First, similarly to (4), it follows that

$$\int_{\hat{S}} \hat{f} d\mu - \nu(\hat{S}) = \int_{\bar{S}} (f_* + \epsilon \chi_{\bar{S}}) d\mu - \nu(\bar{S}) < 0.$$

In particular, this means that $\nu(\bar{S}) > 0$ so $\bar{S}$ is non-empty. Additionally, $\nu(\bar{S}) \leq \mu(S)$. Therefore **if** we can verify that $\hat{f} \in H$ **then** we will contradict maximality of M .

- As argued before, the condition (1) holds for $f + \epsilon \chi_{\hat{S}}$ automatically on sets disjoint from $\hat{S}$, since $f \in H$. By additivity, it suffices to check (1) for any remaining subset $S_0 \subseteq \hat{S}$. (I.e. for a general set E , we can check for both $E \cap \hat{S}$ and $E \setminus \hat{S}$ and add as before.)
- So, suppose for contradiction that $S_0 \subseteq \hat{S}$ violates (1), i.e.

$$\int_{S_0} \hat{f} d\mu > \nu(S_0) + \delta$$

for some positive δ , or equivalently $\text{Def}_\epsilon(S_0) < -\delta$. Then for $k > 1/\delta$, we see that S_k was chosen wrong: we could have used $S_k \cup S_0$ instead, and since they are disjoint we have

$$\text{Def}_\epsilon(S_k \cup S_0) = \text{Def}_\epsilon(S_k) + \text{Def}_\epsilon(S_0) \leq \text{Def}_\epsilon(S_k) - \delta \leq \text{Def}_\epsilon(S_k) - 1/k.$$

This shows that S_k does not obey the approximate-optimality condition (5). This gives the desired contradiction and concludes the proof.

□